Catalytic Reactions and Reactors: II

CENG252
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Rate Limiting Step (P399)

5th edition, chapter 10

- Steady state, heterogeneous reactions;

$$-r_A = r_{a_A} = r_{s_A} = r_{d_A}$$

- Solution Algorithm:
  - Select a mechanism (as molecular or atomic adsorption; single or dual-site reaction: ***L-H mechanism***)
  - Assume a rate limiting step
  - Find the expression for $C_{i\bullet S}$ using the steps that are not rate limiting (occur at equilibrium)
  - Write the site balance
  - Derive the rate law
  - Compare with experimental data

Example 1: Surface Reaction Rate-Limiting (> 75% of all rxns)

Select a mechanism

$$A + S \leftrightarrow A\bullet S$$

Assume a rate limiting step →

$$A\bullet S \leftrightarrow B\bullet S$$

$$B\bullet S \leftrightarrow B + S$$

Find $C_{i\bullet S}$ (steps not rate-limiting occur at high rate/fast)

$$\frac{r_{a/d_A}}{k_A} = \left(P_A C_v - \frac{C_{A\bullet S}}{K_A} \right) \cong 0 \qquad C_{A\bullet S} = K_A P_A C_v$$

$$\frac{r_{a/d_B}}{k_{-B}} = \left(C_{B\bullet S} - K_B P_B C_v \right) \cong 0 \qquad C_{B\bullet S} = K_B P_B C_v$$

Example 1

Select a mechanism $A + S \leftrightarrow A\bullet S$

Assume a rate limiting step → $A\bullet S \leftrightarrow B\bullet S$

$B\bullet S \leftrightarrow B + S$

Find $C_{i\bullet S}$ (steps not rate-limiting occur at high rate/fast)

Write the site balance $C_t = C_v + C_{A\bullet S} + C_{B\bullet S}$

Derive the rate law

$$C_{A\bullet S} = K_A P_A C_v$$

$$r_s = k_s \left(C_{A\bullet S} - \frac{C_{B\bullet S}}{K_s} \right) \neq 0$$

$$C_{B\bullet S} = K_B P_B C_v$$

Example 1

$$C_t = C_v + K_A P_A C_v + K_B P_B C_v$$

$$C_t = C_v\left(1 + K_A P_A + K_B P_B\right)$$

$$C_v = C_t\left(1 + K_A P_A + K_B P_B\right)^{-1}$$

$$r_s = k_s C_v\left(K_A P_A - \frac{K_B}{K_s} P_B\right)$$

$$A + S \leftrightarrow A{\bullet}S$$

rate limiting → $A{\bullet}S \leftrightarrow B{\bullet}S$

$$B{\bullet}S \leftrightarrow B + S$$

$$r_s = \frac{k_s K_A C_t\left(P_A - K_e P_B\right)}{\left(1 + K_A P_A + K_B P_B\right)} = -r_A'$$

(k = $k_s K_A C_t$)

$$K_e = \frac{K_B}{K_A K_s}$$

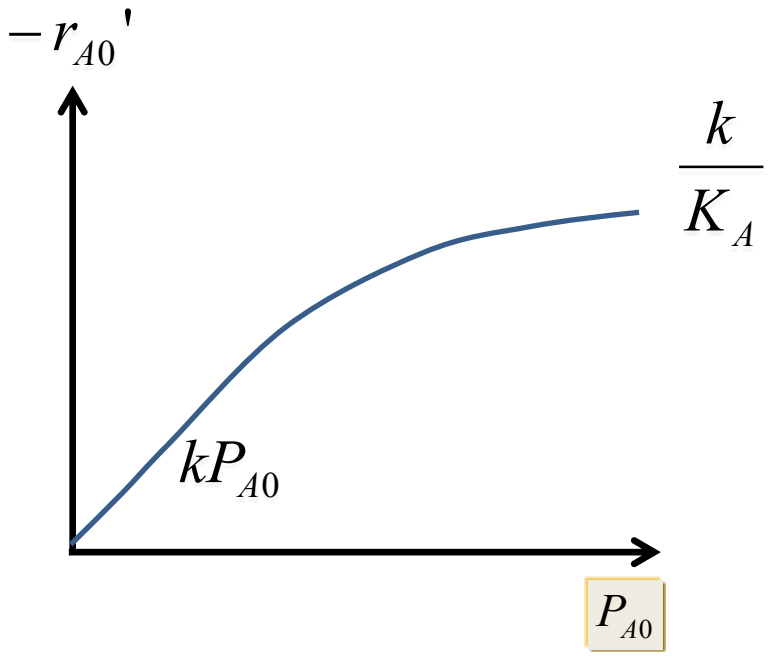
Example 1: The Initial Rate

$$r_s = \frac{\overbrace{k_s K_A C_t}^{k}\left(P_A - K_e P_B\right)}{\left(1 + K_A P_A + K_B P_B\right)} = -r_A'$$



$$-r_{A0}' = \frac{kP_{A0}}{1 + K_A P_{A0}}$$

(small P_{B0})



Example 2: Adsorption Rate-Limiting

another rate limiting step → $A + S \leftrightarrow A{\bullet}S$

$A{\bullet}S \leftrightarrow B{\bullet}S$

$B{\bullet}S \leftrightarrow B + S$

Same mechanism (again)

Find $C_{i\text{-}S}$ (steps not rate-limiting occur at equilibrium)

$$\frac{r_s}{k_s} = \left(C_{A\bullet S} - \frac{C_{B\bullet S}}{K_s} \right) \cong 0 \qquad C_{A\bullet S} = \frac{C_{B\bullet S}}{K_s}$$

$$\frac{r_{a/d_B}}{k_{-B}} = \left(C_{B\bullet S} - K_B P_B C_v \right) \cong 0 \qquad C_{B\bullet S} = K_B P_B C_v$$

Example 2

another rate limiting step → $A + S \leftrightarrow A\bullet S$

Same mechanism (again)

$A\bullet S \leftrightarrow B\bullet S$

$B\bullet S \leftrightarrow B + S$

Find $C_{i\bullet S}$ (steps not rate-limiting occur at equilibrium)

Write the site balance $C_t = C_v + C_{A\bullet S} + C_{B\bullet S}$

Derive the rate law

$$r_{a/d_A} = k_A\left(P_A C_v - \frac{C_{A\bullet S}}{K_A}\right) \neq 0$$

$$C_{A\bullet S} = \frac{C_{B\bullet S}}{K_s}$$

$$C_{B\bullet S} = K_B P_B C_v$$

Example 2

$$C_t = C_v + \frac{K_B P_B C_v}{K_s} + K_B P_B C_v$$

rate limiting →

$$A + S \leftrightarrow A{\bullet}S$$

$$A{\bullet}S \leftrightarrow B{\bullet}S$$

$$B{\bullet}S \leftrightarrow B + S$$

$$C_t = C_v\left(1 + \frac{K_B P_B}{K_s} + K_B P_B\right)$$

$$C_v = C_t\left[1 + \left(K_B + K_e K_A\right)P_B\right]^{-1}$$

$$r_s = \frac{k_A C_t\left(P_A - \frac{P_B}{K_e}\right)}{1 + P_B\left(K_B + K_e K_B\right)}$$

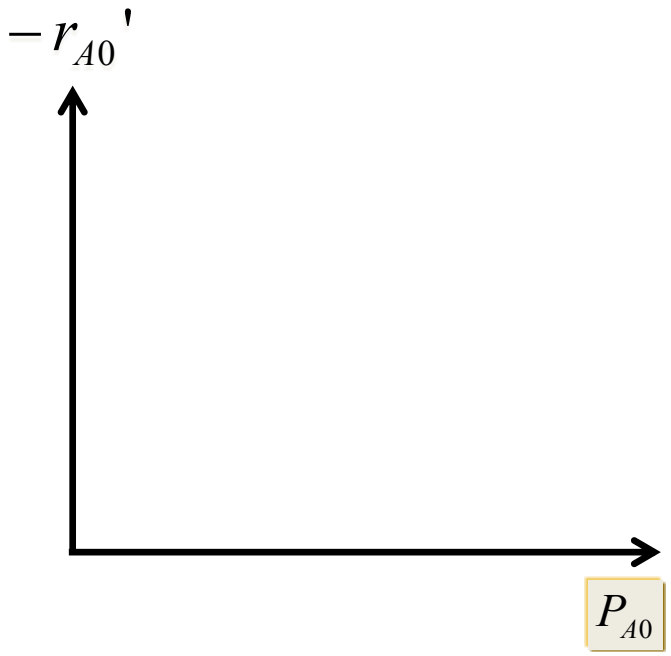
$$r_{a/d_A} = k_A C_v\left(P_A - \frac{K_B}{K_s K_A} P_B\right)$$

$$K_e = \frac{K_A K_s}{K_B}$$

Example 2: The Initial Rate

(small P_{B0})

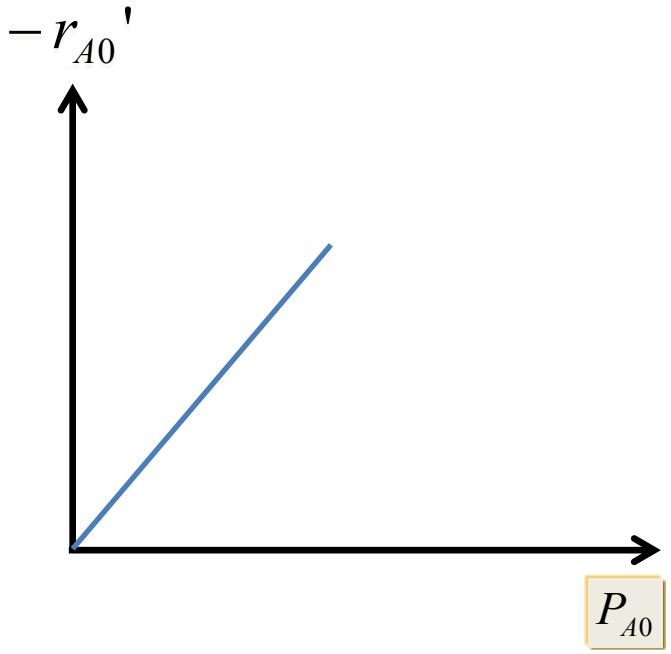
$$r_s = \frac{k_A C_t \left(P_A - \frac{P_B}{K_e} \right)}{1 + P_B \left(K_B + K_e K_B \right)}$$



Example 2: The Initial Rate

(small P_{B0})

$$r_s = \frac{k_A C_t \left(P_A - \frac{P_B}{K_e} \right)}{1 + P_B (K_B + K_e K_B)}$$



L-H Analysis Protocol

- Protocol for developing rate laws for heterogeneous catalytic reactions using Langmuir-Hinshelwood kinetic models:
  1. Propose a mechanism
  2. Write net rate laws for each step in mechanism
  3. Identify (select) a rate-limiting step
  4. Apply *pseudo-steady state hypothesis*
  5. Eliminate adsorbed species concentrations
  6. Write total site balance, solve for vacant sites
  7. Substitute C_v and group constants

Propose Mechanism/Rate Laws

$$A + S \leftrightarrow A{\bullet}S$$

$$W + S \leftrightarrow W{\bullet}S$$

$$W{\bullet}S + A{\bullet}S \leftrightarrow B{\bullet}S + S$$

$$B{\bullet}S \leftrightarrow B + S$$

Propose Mechanism/Rate Laws

$$A + S \leftrightarrow A{\bullet}S \qquad r_A = k_A\left(C_A C_v - \frac{C_{A\bullet S}}{K_A}\right)$$

$$W + S \leftrightarrow W{\bullet}S \qquad r_W = k_W\left(C_W C_v - \frac{C_{W\bullet S}}{K_W}\right)$$

$$W{\bullet}S + A{\bullet}S \leftrightarrow B{\bullet}S + S \qquad r_s = k_s\left(C_{A\bullet S} C_{W\bullet S} - \frac{C_{B\bullet S} C_v}{K_S}\right)$$

$$B{\bullet}S \leftrightarrow B + S \qquad r_B = k_B\left(\frac{C_{B\bullet S}}{K_B} - C_B C_v\right)$$

Rate-Limiting Step & PSSH

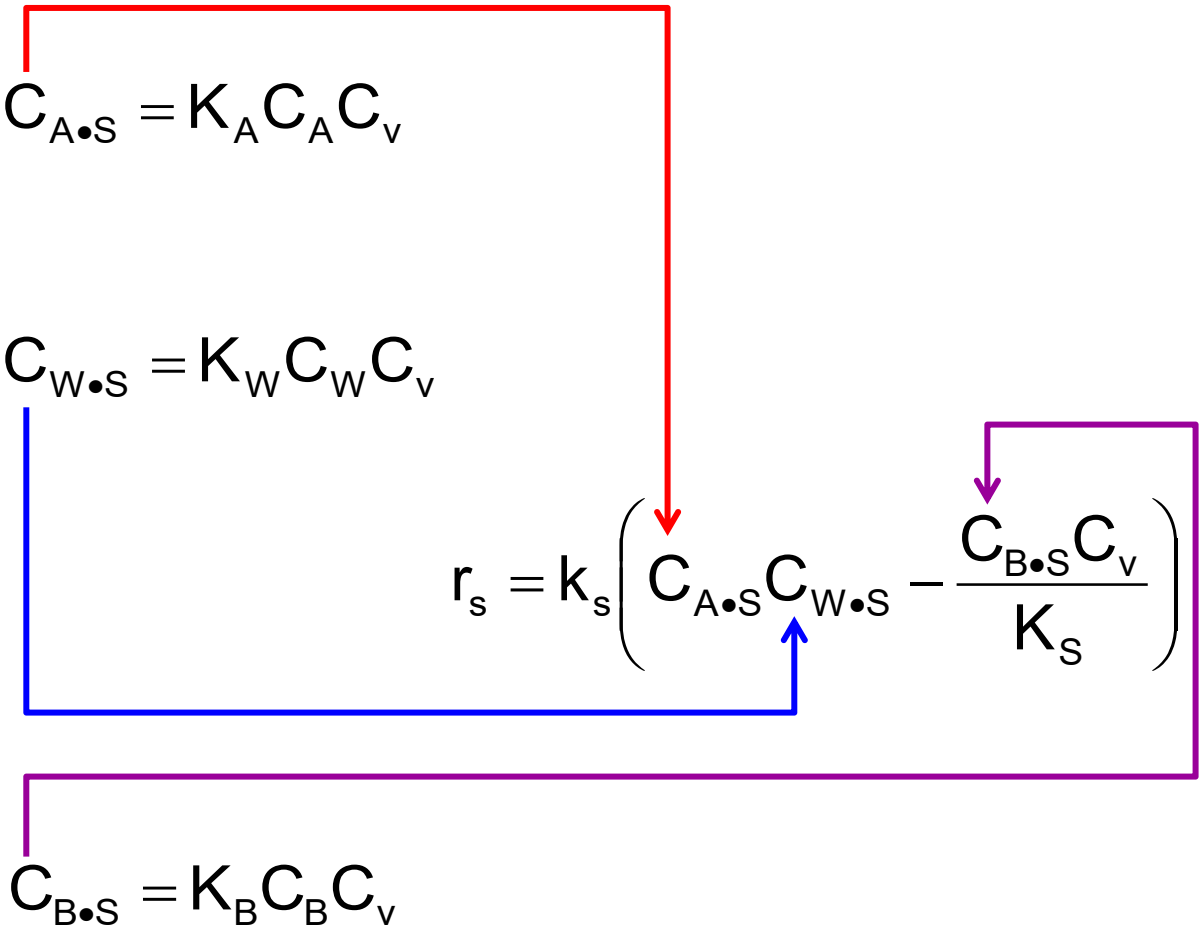
$$A + S \leftrightarrow A\bullet S \qquad C_{A\bullet S} = K_A C_A C_v \qquad \frac{r_A}{k_A} = \left(C_A C_v - \frac{C_{A\bullet S}}{K_A} \right) \cong 0$$

$$W + S \leftrightarrow W\bullet S \qquad C_{W\bullet S} = K_W C_W C_v \qquad \frac{r_W}{k_W} = \left(C_W C_v - \frac{C_{W\bullet S}}{K_W} \right) \cong 0$$

$$W\bullet S + A\bullet S \leftrightarrow B\bullet S + S \qquad r_s = k_s \left(C_{A\bullet S} C_{W\bullet S} - \frac{C_{B\bullet S} C_v}{K_S} \right)$$

$$B\bullet S \leftrightarrow B + S \qquad C_{B\bullet S} = K_B C_B C_v \qquad \frac{r_B}{k_B} = \left(\frac{C_{B\bullet S}}{K_B} - C_B C_v \right) \cong 0$$

Eliminate X•S Concentrations



Eliminate X•S Concentrations

$$r_s = k'\left(C_A C_W - \frac{C_B}{K_P} \right) \qquad K_P = \frac{K_S K_A K_W}{K_B} \qquad k' = k_S K_A K_W C_v^2$$

$$r_s = k_s\left(C_{A\bullet S} C_{W\bullet S} - \frac{C_{B\bullet S} C_v}{K_S} \right)$$

Perform Site Balance

$$r_s = k'\left(C_A C_W - \frac{C_B}{K_P}\right) \qquad K_P = \frac{K_S K_A K_W}{K_B} \qquad k' = k_S K_A K_W C_v^2$$

$$C_T = C_v + C_{A\bullet S} + C_{W\bullet S} + C_{B\bullet S}$$

$$C_T = C_v + K_A C_v C_A + K_W C_v C_W + K_B C_v C_B$$

$$C_T = C_v\left(1 + K_A C_A + K_W C_W + K_B C_B\right)$$

$$C_v = C_T\left(1 + K_A C_A + K_W C_W + K_B C_B\right)^{-1}$$

recall:

$$C_{A\bullet S} = K_A C_A C_v$$

$$C_{W\bullet S} = K_W C_W C_v$$

$$C_{B\bullet S} = K_B C_B C_v$$

Perform Site Balance

$$r_s = k'\left(C_A C_W - \frac{C_B}{K_P}\right) \qquad K_P = \frac{K_S K_A K_W}{K_B} \qquad k' = k_S K_A K_W C_v^2$$

$$r_s = \frac{k_S K_A K_W C_T^2 \left(C_A C_W - \dfrac{K_B}{K_S K_A K_W} C_B\right)}{\left(1 + K_A C_A + K_W C_W + K_B C_B\right)^2}$$

Rate law for mechanism with surface reaction as limiting kinetics

$$C_v = C_T \left(1 + K_A C_A + K_W C_W + K_B C_B\right)^{-1}$$

Propose Mechanism/Rate Laws

$A + S \leftrightarrow A \bullet S$

$$r_A = k_A \left(C_A C_v - \frac{C_{A \bullet S}}{K_A} \right)$$

$W + S \leftrightarrow W \bullet S$

$$r_W = k_W \left(C_W C_v - \frac{C_{W \bullet S}}{K_W} \right)$$

$W \bullet S + A \bullet S \leftrightarrow B \bullet S + S$

$$r_s = k_s \left(C_{A \bullet S} C_{W \bullet S} - \frac{C_{B \bullet S} C_v}{K_S} \right)$$

$B \bullet S \leftrightarrow B + S$

$$r_B = k_B \left(\frac{C_{B \bullet S}}{K_B} - C_B C_v \right)$$

Rate-Limiting Step & PSSH

$$A + S \leftrightarrow A{\bullet}S \qquad r_A = k_A\left(C_A C_v - \frac{C_{A\bullet S}}{K_A}\right)$$

$$W + S \leftrightarrow W{\bullet}S \qquad C_{W\bullet S} = K_W C_W C_v \qquad \frac{r_W}{k_W} = \left(C_W C_v - \frac{C_{W\bullet S}}{K_W}\right) \cong 0$$

$$W{\bullet}S + A{\bullet}S \leftrightarrow B{\bullet}S + S \qquad C_{A\bullet S} = C_v C_{B\bullet S} / K_s C_{W\bullet S} \qquad \frac{r_s}{k_s} = \left(C_{A\bullet S} C_{W\bullet S} - \frac{C_{B\bullet S} C_v}{K_S}\right) \cong 0$$

$$B{\bullet}S \leftrightarrow B + S \qquad C_{B\bullet S} = K_B C_B C_v \qquad \frac{r_B}{k_B} = \left(\frac{C_{B\bullet S}}{K_B} - C_B C_v\right) \cong 0$$

$C_{X\bullet S}$, Site Balance, Substitute

A + S ↔ A•S

$$r_A = k_A\left(C_A C_v - \frac{C_{A\bullet S}}{K_A}\right)$$

W + S ↔ W•S

$$C_{W\bullet S} = K_W C_W C_v$$

site balance unchanged

$$C_v = \frac{C_T}{1 + K_A C_A + K_W C_W + K_B C_B}$$

W•S + A•S ↔ B•S + S

$$C_{A\bullet S} = C_v C_B / K_s C_{W\bullet S}$$

B•S ↔ B + S

$$C_{B\bullet S} = K_B C_B C_v$$

$$r_A = \frac{k_A C_T\left(C_A - \frac{C_B}{C_W K_P}\right)}{1 + K_W C_W + K_B C_B + \frac{C_B}{K_p C_W}}$$